

Haniyeh Jalayeri

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SUMMARY

Full stack developer with hands-on experience building and maintaining production features in Python (FastAPI), PostgreSQL, React, TypeScript, Node.js, and Remix. Skilled at debugging, root-cause analysis, and writing test cases that keep code reliable. Also builds AI applications, including a retrieval-augmented generation (RAG) system and tool-calling LLM agents. Quick to learn unfamiliar codebases.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C#, SQL

Backend: FastAPI, Pydantic, SQLAlchemy (async), REST APIs, Node.js, .NET, Entity Framework; PostgreSQL, SQL Server, Relational Database Design

Frontend: React, Remix, HTML, CSS, Tailwind CSS, Material UI

AI & Data: RAG, LLM Agents, Vector Databases (Qdrant), Embeddings (sentence-transformers), Anthropic Claude API (tool use), OpenAI Agents SDK, Prompt Design

Testing & Practices: Test Case Design, Debugging, Root-Cause Analysis, Data Validation, Code Review, Agile

Cloud & Tools: Microsoft Azure, Docker, Docker Compose, Git, GitHub, Bitbucket, Jira, Confluence

WORK EXPERIENCE

Full Stack Developer Co-op | MEDIC / Mohawk College IDEAWORKS May 2026 – Present

- Build and maintain application features across the stack using Python, FastAPI, PostgreSQL, React, TypeScript, Node.js, and Remix.
- Develop multi-factor authentication (MFA), including TOTP-based verification and secure secret storage.
- Write test cases to verify new features and bug fixes and to catch regressions before code is merged.
- Diagnose and fix bugs across the frontend, REST APIs, database, authentication, and development environment using systematic debugging and root-cause analysis.
- Build and integrate REST APIs and backend services that validate, process, store, and retrieve application data.
- Collaborate with developers using Git branches, pull requests, code reviews, Docker, and Agile practices.

Math Tutor | Mohawk College May 2025 – Apr 2026

- Explained technical and analytical concepts clearly to students with varying levels of experience.

Postdoctoral Fellow / Researcher | Université de Sherbrooke; University of Tehran, Iran Sep 2018 – Feb 2022

- Analyzed large datasets, validated results, and prepared technical reports; managed data-driven research projects for multidisciplinary teams.

PROJECTS

World Cup 2026 AI Analytics (RAG + Agent) | Python, FastAPI, Qdrant, PostgreSQL, Claude API, Docker

- Built a RAG backend that answers natural-language questions about 48 teams, 104 matches, and 1,248 players by embedding text summaries generated from PostgreSQL records with sentence-transformers and retrieving them from a Qdrant vector store using cosine similarity and metadata filters.
- Designed an indexing pipeline with deterministic, hash-based point IDs so re-indexing updates existing vectors instead of duplicating them.
- Built a Claude tool-use agent that routes each question to one of five fixed, parameterized database queries, the vector knowledge base, or live web search (Tavily), with a capped tool-call loop; avoided text-to-SQL to keep queries safe and predictable.
- Modeled the tournament data in async SQLAlchemy, seeded it from CSV files, and exposed teams, matches, RAG, and agent endpoints through a FastAPI REST API; containerized the API, PostgreSQL, and Qdrant with Docker Compose.

FlexCare – AI Recovery Assistant | React, Python, FastAPI, Gemini API, Railtracks, Tailwind CSS

- Built a full-stack AI recovery assistant at GenAI Genesis Hackathon 2026 with a team, coordinating a multi-agent pipeline (assessment, safety, recovery or referral) behind a React frontend and FastAPI backend.

AI Healthcare Triage Agent (MedBot) | Python, OpenAI Agents SDK

- Developed an LLM-powered triage application using the OpenAI Agents SDK with custom tools and structured workflows.
- Designed typed tool interfaces and validation logic to improve reliability, and implemented Python ranking logic over structured data.

EDUCATION

Advanced Diploma, Computer Systems Technology – Software Development | Mohawk College, Hamilton, ON | GPA: 93%

Ph.D., Mining Engineering | Shahrood University of Technology, Iran

CERTIFICATIONS, AWARDS & HONOURS

- Microsoft Certified: Azure Fundamentals (AZ-900), March 2026
- Dean's Honours List (4 terms); Morris Epstein Entrance Scholarship (2025); Zonta Club of Hamilton 1 Bursary (2025)